

# Noah Lin Cruz *Applied AI Engineer*

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## SUMMARY

Applied AI Engineer specializing in LLMs, RAG, AI Agents, and machine learning systems. Experienced in developing AI prototypes using Python, PyTorch, and TensorFlow, with knowledge of model evaluation, prompt engineering, vector search, and AI application development. Strong mathematical foundation in optimization, statistics, and machine learning.

## AI EXPERIENCE

**AI Development Researcher** 06/2026 – Present  
*Independent*

- Developed machine learning and AI agent systems using Python, PyTorch, TensorFlow, and modern web technologies.
- Designed and evaluated machine learning experiments involving prompt optimization, model evaluation, and performance benchmarking, achieving approximately 5–10% improvements in response quality through iterative testing and refinement.
- Developed prototype AI agent workflows integrating tool usage, reasoning pipelines, and automation strategies, improving simulated task completion efficiency by approximately 10–15%.
- Built retrieval-augmented generation (RAG) prototypes using embedding models and vector databases, reducing information retrieval latency by approximately 15–20%.
- Created AI-powered web application prototypes using TypeScript and Next.js, integrating machine learning APIs to demonstrate practical AI application scenarios.
- Researched transformer architectures, LLM application frameworks, and AI agent methodologies while documenting findings and evaluating their potential for real-world AI systems

## EDUCATION

**Palawan State University** 08/2024 – 05/2026  
*Master of Arts in Education, Mathematics* Puerto Princesa City, Philippines

- GPA: 3.0 (GWA 2.0)
- Achievements: Conducted a master's research project under faculty supervision.
- Coursework: Advanced Research Methods, Advanced Statistics, Philosophy of Education, Seminar in Modern Mathematics, Foundations of Modern Mathematics, Mathematical Analysis, Elementary Statistics and Probability, Sets and Modern Algebra, Ordinary Differential Equations, Teaching of Elementary Mathematics, Thesis/Research Requirement, Comprehensive Examination

**Palawan State University** 08/2020 – 05/2024  
*Bachelor of Secondary Education, Mathematics* Puerto Princesa City, Philippines

- GPA: 3.25 (GWA 1.75)
- Achievements: Successfully completed undergraduate mathematics coursework with a focus on analytical problem-solving.
- Coursework: College Algebra, Trigonometry, Calculus, Geometry, Linear Algebra, Statistics and Probability, Number Theory, Abstract Algebra, Mathematics Teaching Methods

## TECHNICAL SKILLS

|                                                                                                                                                                                                                      |                                                                                                     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| <b>AI / Machine Learning</b><br>Large Language Models(LLMs), Transformer Architectures, Retrieval-Augmented Generation (RAG), AI Agents, Reinforcement Learning, Deep Learning, Model Evaluation, Prompt Engineering | <b>Frameworks &amp; Libraries</b><br>Python, PyTorch, TensorFlow, scikit-learn, NumPy, Pandas       |
| <b>AI Systems</b><br>Vector Search, Knowledge Retrieval, Model Fine-tuning, Distributed Training, Inference Optimization                                                                                             | <b>Software Engineering</b><br>TypeScript, Next.js, HTML/CSS, REST APIs, Full-stack AI Applications |
| <b>Mathematics</b><br>Linear Algebra, Probability & Statistics, Optimization, Calculus, Mathematical Modeling                                                                                                        |                                                                                                     |

## SELECTED AI PROJECTS

### **Autonomous AI Agent Framework**

*Python / PyTorch / Reinforcement Learning*

- Designed an AI agent architecture combining reinforcement learning concepts with intelligent decision-making workflows.
- Evaluated agent behavior in simulated environments and optimized decision efficiency.

### **Domain-Specific LLM Research System**

*Python / Transformers / RAG*

- Built retrieval-augmented generation workflows using domain-specific scientific data.
- Improved information retrieval performance through optimized retrieval strategies and prompt refinement.

## RESEARCH INTERESTS

Machine Learning Systems, Large Language Models, AI Agents, Reinforcement Learning, Retrieval-Augmented Generation, and Applied Artificial Intelligence.